✨ Features of Python

Python is one of the most popular and beginner-friendly programming languages.
It is widely used in web development, data science, artificial intelligence, and automation.
Let's look at its main features 👇

1️⃣ Easy to Learn and Read

- Python has a simple and clean syntax — looks almost like English.
- Easy for beginners to learn and understand.
  Example:

```
print("Hello, World!")
```

2️⃣ Interpreted Language

- Python code is executed **line by line**.
- You don't need to compile it before running.
- This makes debugging easier.

3️⃣ Dynamically Typed

- You **don't need to declare data types** before using variables.
- Python automatically detects the type at runtime.
  Example:

```
a = 10      # integer
a = "Hello" # now string
```

4️⃣ Object-Oriented Language

- Python supports **Object-Oriented Programming (OOP)** — using classes and objects.
- Makes code reusable and easier to organize.

5️⃣ Free and Open Source

- Python is **free to download and use**.
- It's **open source**, meaning anyone can view or modify its code.

6️⃣ Large Standard Library

- Python comes with a **huge collection of built-in modules** (like `math`, `datetime`, `os`, etc.).
- Saves time because you don't have to write everything from scratch.

7️⃣ Platform Independent

- Python programs can run on **Windows, macOS, Linux**, etc. without changes.
- Just install Python and run the same code anywhere.

8️⃣ Extensible and Embeddable

- You can **add code from other languages** like C or C++ inside Python.
- You can also **embed Python inside other applications**.

9️⃣ Portable

- Write your program once and run it anywhere — **“Write once, run anywhere.”**

🔟 Huge Community Support

- Python has a **large and active community** of developers.
- Easy to find help, tutorials, and ready-made libraries online.

1️⃣1️⃣ Supports Multiple Programming Paradigms

- Python supports:
    - **Procedural programming** (using functions)
    - **Object-oriented programming** (using classes and objects)
    - **Functional programming** (using `lambda`, `map`, `filter`, etc.)

1️⃣2️⃣ High-Level Language

- You don’t need to worry about low-level details like memory management.
- Python handles those automatically.

1️⃣3️⃣ Embeds with Other Technologies

- Python can connect easily with **databases, APIs, and web frameworks** like Django or Flask.

- Useful for developing full applications.

✅ Summary

Feature	Description
Easy to Learn	Simple, readable syntax
Interpreted	Executes line by line
Dynamically Typed	No need to declare variable type
Object-Oriented	Supports classes & objects
Free & Open Source	Available for everyone
Large Library	Comes with many built-in modules
Cross-Platform	Works on all operating systems
Community Support	Huge online community
Portable	Write once, run anywhere
Multi-Paradigm	Supports multiple coding styles